```
#include <Adafruit_GFX.h>
#include <Adafruit_PCD8544.h>
#include <SPI.h>
Adafruit_PCD8544 display= Adafruit_PCD8544 (8,9,10,11,12);
int val;
int tempPin = A0;
int fan = 5;
int temp;
int tempMin = 20;
int tempMax = 35;
int fanSpeed;
int fanLCD;

void setup() {
display.begin();  // put your setup code here, to run once:
display.setContrast(40);
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(BLACK,WHITE);
display.setCursor(10,8);
display.print("TEMPERATURE");
display.display();
display.setCursor(53,20);
display.print("degC");
display.display();
pinMode(fan,OUTPUT);
pinMode(tempPin,INPUT);
}
```

```
void loop() {
val = analogRead(tempPin);
float mv = (val/1024.0)*5000;
float temp = mv/10;
display.setCursor(18,20);
display.print(temp);
display.display();
delay(1000);
 if(temp < tempMin) {   // if temp is lower than minimum temp
      fanSpeed = 0;      // fan is not spinning
      digitalWrite(fan, LOW);
   }
   if((temp >= tempMin) && (temp <= tempMax)) {  // if temperature is higher than minimum temp
      fanSpeed = map(temp, tempMin, tempMax, 32, 255); // the actual speed of fan
      fanLCD = map(temp, tempMin, tempMax, 0, 100);  // speed of fan to display on LCD
      analogWrite(fan, fanSpeed);  // spin the fan at the fanSpeed speed
   }
   display.setCursor(0,30);   // move cursor to next line
  display.print("FANS: ");
   display.print(fanLCD);    // display the fan speed
   display.print("%");
   delay(200);
   }
```

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ABSTRACT

The aim of this research work is to design and construction of microcontroller based fan speed control by using temperature sensor. The required components for this research are microcontroller (ATmega328), temperature sensor (LM-35), Nokia 5110 Graphic (LCD) module, power supply, motor driver, the direct current (DC) motor and switches. The temperature sensor senses the surrounding temperature and outputs analog voltage. This analog voltage is fed to the analog input of microcontroller. Microcontroller converts this analog voltage to digital data and outputs Pulse Width Modulation (PWM) signal according to received digital data. The PWM signal generated from microcontroller activates fan driver. The fan's speed will change according to the temperature of the room or surrounding.

CHAPTER I

INTRODUCTION

Electric fan is one of the most popular electrical devices due to its cost effectiveness and lower power consumption advantages. It is a common circuit and widely used in many applications. It is also one of the most sensible solutions to offer a comfortable and energy efficient. In fact, the fan has been long used and still available in the market. Fan speed needs to be manually controlled every time.

Nowadays, the humankind is moving towards the new technologies by replaying the manual operations to automatic controlled devices. This research will convert the manual fan into automatic fans. The automatic fans will change its speed according to the temperature in the room. Therefore, the people will control easily the speed of fan by using this automatic fan. An automatic temperature control system technology is needed for the controlling purpose in the fan speed according to the temperature changes.

Variable speed DC drives have become widespread in home appliance area. The DC motor is usually employed in many applications as it gives variable speed and load characteristics due to its ease of controllability. Microcontroller provides a suitable means of control and interface functions. The DC motor with chopper has higher efficiency as there are no copper and iron losses. The main advantage of PWM circuit over a resistive power controller is that in a resistive controller system most of the power is wasted in series resistor in the form of heat. A reasonable number of works have been found in the literature regarding the control of DC drives. The control algorithms are stored and implemented by the microprocessor.

This research results in the automation of fans where there will no need to switch the regulators up and down to change the speed of fan. So by the usage of microcontroller the fans can automatically change is Revolution per Minute (RPM) by the use of heat and temperature sensor. The sensor will sense the

temperature and the fan speed will be controlled using the Pulse Width Modulation (PWM) and Microcontroller. The microcontroller can be used to control the fan speed according to the temperature of the room.

In this thesis designing of Microcontroller based temperature control for DC motor has been proposed. The microcontroller system is equipped with a graphic LCD display and a temperature sensor (LM-35). Software is written to show the value of temperature and motor speed on the monitor. The hardware structure of the system consists of ATmega328, a DC motor, temperature sensor (LM-35) and LCD display. The software used to write programs is based on the Arduino IDE (Integrated Development Environment). The block diagram of fan speed control system is shown in Fig. (1.1).

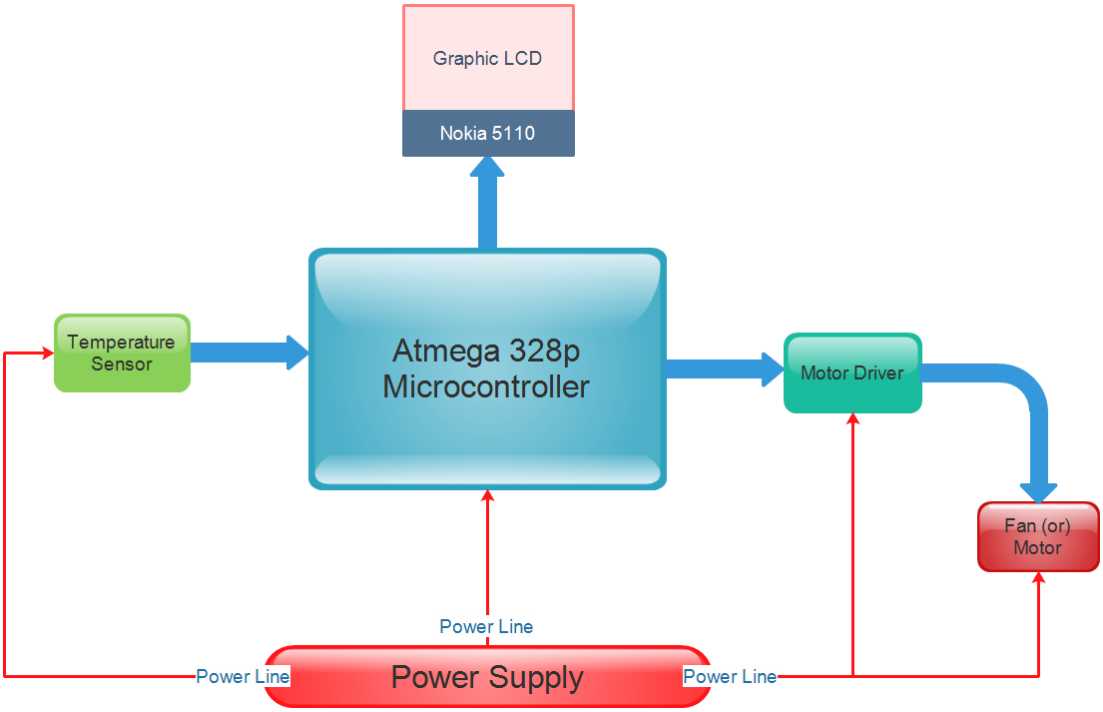


Fig. (1.1) Block diagram of fan speed control system

CHAPTER II

BACKGROUND THEORY OF ELECTRONIC DEVICES

The microcontroller-based fan speed control system consists of ATmega328 microcontroller, temperature sensor, DC motor, graphic LCD module and other electric components. The general features of these devices will be explained in this chapter.

2.1 Microcontroller System

A microcontroller is a single chip computer. Micro suggests that the device is small, and controller suggests that the device can be used in control applications. Another term used for microcontrollers is embedded controller, since most of the microcontrollers are built into (or embedded in) the devices they control. The simplest microcontroller architecture consists of a microprocessor, memory, and input/ output. The microprocessor consists of a central processing unit (CPU) and the control unit (CU). The CPU is the brain of the microcontroller and this is where all of the arithmetic and logic operations are performed. The CU controls the internal operations of the microprocessor and sends out control signals to other parts of the microcontroller to carry out the required instructions. Memory is an important part of a microcontroller system. Depending upon the type used memories can be classified into two groups: program memory and data memory. Program memory stores the program written by the programmer and this memory is usually non-volatile, that is data is not lost after the removal of power. Data memory is where the temporary data used in a program are stored and this memory is usually volatile, that is data is lost after the removal of power.

There are basically five types of memories. They are RAM, ROM, EPROM, EEPROM and Flash EEPROM. RAM means Random Access Memory. It is a general-purpose memory which usually stores the user data used in a program. RAM is volatile, that is data is lost after the removal of power. ROM is Read Only Memory. This type of memory usually holds program or fixed user data.

EPROM is Erasable Programmable Read Only Memory. This is similar to ROM, but the EPROM can be programmed using a suitable programming device. EEPROM is Electrically Erasable Programmable Read Only Memory, which is a non-volatile memory. These memories can be erased and also be programmed under program control. EEPROMs are used to save configuration information, maximum and minimum values, identification data, etc. Flash EEPROM is non-volatile and is usually very fast. The data is erased and then re-programmed using a programming device.

2.2 The Hardware Features of Microcontroller

Most microcontrollers operate with the standard logic voltage of +5 V. Some microcontrollers can operate at as low as +2.7 V and some will tolerate +6 V without any problems. A voltage regulator circuit is usually used to obtain the required power supply voltage when the device is to be operated from a mains adaptor or batteries.

The microcontroller requires a clock (or an oscillator) to operate. The clock is usually provided by connecting external timing devices to the microcontroller. Most microcontrollers will generate clock signals when a crystal and two small capacitors are connected. Some will operate with resonators or external resistor-capacitor pair. Some microcontrollers have built-in timing circuits and they do not require any external timing components. The microcontroller is actually operated at a clock rate which is a quarter of the actual oscillator frequency.

Timers are important parts of any microcontroller. A timer is basically a counter which is driven either from an external clock pulse or from the internal oscillator of the microcontroller. It is typical to have at least one timer in every microcontroller. Some microcontrollers may have two, three, or even more timers where some of the timers can be cascaded for longer counts.

Most microcontrollers have at least one watchdog facility. The watchdog is basically a timer which is refreshed by the user program and a reset occurs if the program fails to refresh the watchdog.

A reset input is used to reset a microcontroller. An external reset action is usually achieved by connecting a push-button switch to the reset input such that the microcontroller can be reset when the switch is pressed.

Interrupts are very important concepts in microcontrollers. An interrupt causes the microcontroller to respond to external and internal (e.g. a timer) events very quickly.

An analogue-to-digital converter (ADC) is used to convert an analogue signal such as voltage to a digital form so that it can be read by a microcontroller. Some microcontrollers have built-in ADC.

EEPROM type data memory is also very common in many microcontrollers. The advantage of an EEPROM memory is that the programmer can store non-volatile data in such a memory, and can also change this data whenever required.

Analogue comparators are used where it is required to compare two analogue voltages. Although these circuits are implemented in most high-end PIC microcontrollers they are not common in other microcontrollers.

2.3 Atmel ATmega328 Microcontroller IC

A microcontroller is a tiny computer that contains a processor that executes instructions, and various types of memory to hold data and instructions from sketch, and various ways to send and receive data. The ATmega328 is shown in Fig. (2.1). When looking at the IC in the figure, notice that pin number 1 is at the bottom-left of the IC and is marked by a small dot. The ATmega328 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture. The ATmega328 achieves throughputs approaching 1 MIPS per MHz allowing the system designed to optimize power consumption versus processing speed. Table (2.1) lists the Arduino pins on the left and the matching ATmega328 pins on the right.

2.4 Arduino Software

The software used to write programs for Arduino is called the Arduino IDE (Integrated Development Environment). The Arduino IDE is based on another open source programming language and program called Processing used for programming images, animations and computer interactions.

Even the Arduino coding language is molded after the Processing language. The Arduino language is based on easy to use commands and every time press the upload button in the Arduino IDE, and it will translate program into C code. So the Arduino board can understand the program. The software consists of a standard programming language compiler and a boot loader that executes on the microcontroller. Arduino IDE processing can see the Fig. (2.2).

2.5 Temperature Sensor (LM-35)

The LM-35 series are precision integrated-circuit temperature sensors, with an output voltage linearly proportional to the Centigrade temperature. Thus the LM-35 has an advantage over linear temperature sensors calibrated in Kelvin, as the user is not required to subtract a large constant voltage from the output to obtain convenient Centigrade scaling. The scale factor of LM-35 is 10 mV per degree Celsius. As the temperature increases, the output voltage increases. The LM-35 does not require any external calibration or trimming to provide typical accuracies of $\pm 0.25℃$ at room temperature and $\pm 0.75℃$ over a full $-55℃$ to $+150℃$ temperature range. The low output impedance, linear output, and precise inherent calibration of the LM-35 make interfacing to readout or control circuitry especially easy. The device is used with single power supplies, or with plus and minus supplies. As the LM-35 draws only 60 μA from the supply, it has very low self-heating of less than $0.1℃$ in still air. The LM-35 is rated to operate over a $-55℃$ to $+150℃$ temperature range, while the LM-35C is rated for a $-40℃$ to $+110℃$ range ($-10°$ with improved accuracy). Photograph and plastic package of LM-35 are shown in Fig. (2.3).

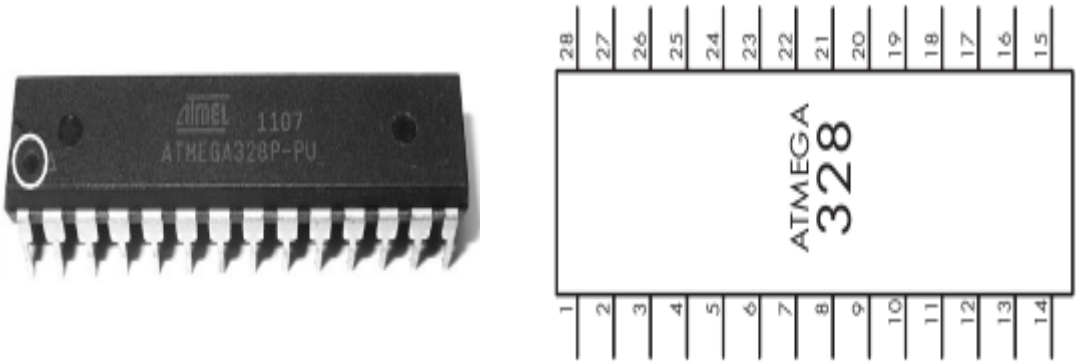


Fig. (2.1) ATmega328

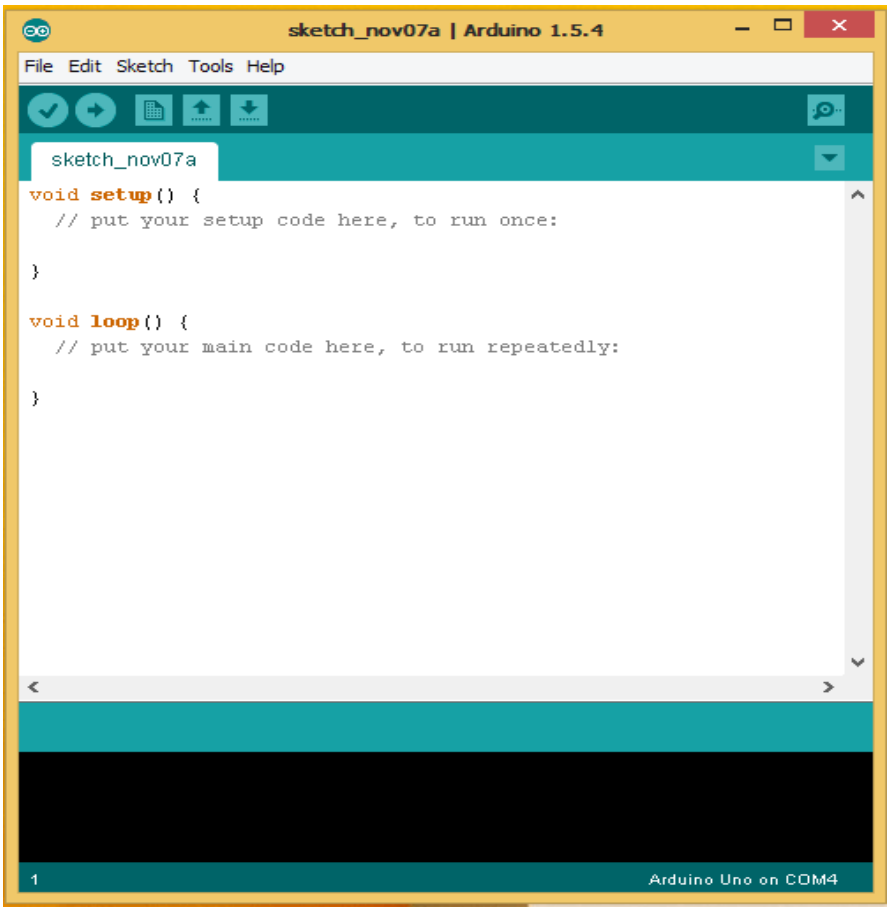


Fig. (2.2) Arduino IDE

Table (2.1) Pins for ATmega328

Arduino Pin Name	ATmega328 Pin
RST	1
RX/D0	2
TX/D1	3
D2	4
D3	5
D4	6
(5 V only)	7
GND	8
D5	11
D6	12
D7	13
D8	14
D9	15
D10	16
D11	17
D12	18
D13	19
(5 V only)	20
AREF	21
GND	22
A0	23
A1	24
A2	25
A3	26
A4	27
A5	28

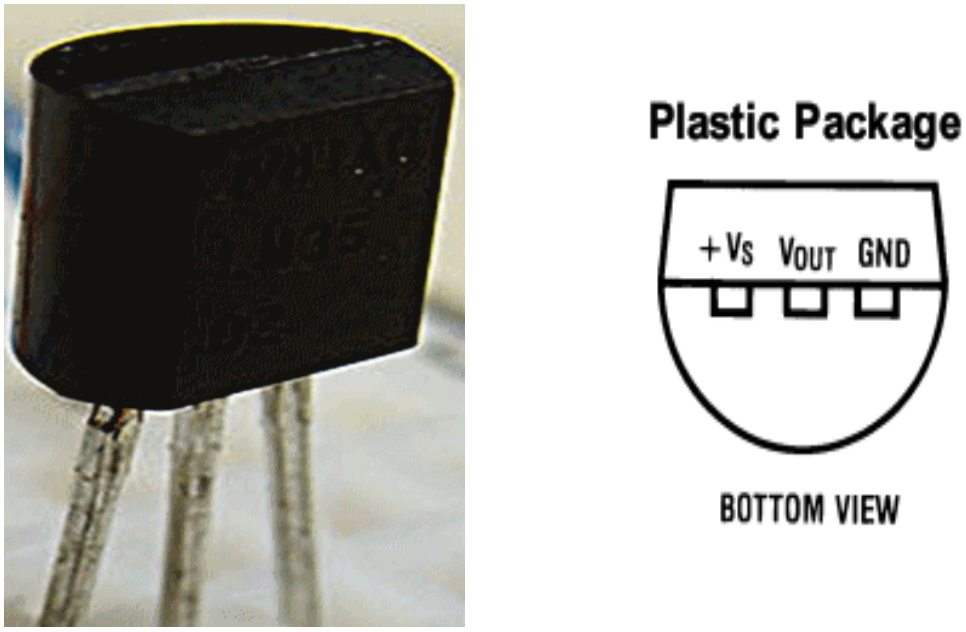


Fig. (2.3) Photograph and plastic package of LM-35

2.6 DC Motor

The direct current (DC) motor is one of the first machines devised to convert electrical energy to mechanical energy. Its origin can be traced to machines conceived and tested by Michael Faraday, the experimenter who formulated the fundamental concepts of electromagnetism. When a current carrying conductor is placed in a magnetic field, it experiences a torque and has a tendency to move. This is known as motoring action. If the direction of electric current in the wire is reversed, the direction of rotation also reversed. When magnetic field and electric field interact, they produce a mechanical force and based on that the working principle of DC motor established.

If more current is passed through the winding (turns of wire), then more force (torque) is obtained. In effect, two magnetic fields interacting cause movement: the magnetic field from the rotor and the magnetic field from the stators attract each other. This becomes the basis of both AC and DC motor design. The direction of rotation of this motor is given by Fleming's left hand rule, which states that if the index finger, middle finger and thumb of your left hand are extended mutually perpendicular to each other and if the index finger represents the direction of magnetic field, middle finger indicates the direction of electric current, then the thumb represents the direction in which force is experienced by the shaft of the DC motor. DC motor speeds can easily be varied, therefore they are utilized in applications where speed control, servo control or positioning needs exist and home appliance.

2.7 Pulse Width Modulation (PWM)

Pulse Width Modulation is a way where the cyclic sequence pulses with changed according to the baseband signal. The purpose of Pulse Width Modulation is to manage power delivery, particularly to mechanical-electrical devices. The on-off behavior changes the typical power of the signal. Output signal alternates between on and off among a specific amount of time. When signal switches between on and off, faster than the load, then the load isn't

overdone with the switching. Another use of PWM is to cipher data for transmission. In PWM, the pulse width is proportional to the amplitude of the signal. By changing the duty cycle of the pulse, the speed of the fan will be controlled. The duty cycle can be outlined because the quantity of time in a very explicit amount throughout that the pulse is active or high. A low frequency pulse width modulation (PWM) signal, usually in the range of about 30 Hz, whose duty cycle is varied to adjust the fan's speed. An inexpensive single, small pass transistor can be used in this research. The speed is formed slow, medium, and fast in no time and 0 by having totally different duty cycles.

2.8 Nokia 5110 Graphic LCD screen

The Nokia 5110 is a basic graphic LCD screen for lots of applications. It was originally interfaced to be used a cell phone screen. So, it is capable of displaying alphanumeric characters, a bitmap image. All this is possible because of its (84×48) monochrome pixels. At the heart of the module is a powerful single-chip low power CMOS LCD driver controller form Philips-PCD8544. The PCD8544 LCD driver has a built-in 504 bytes Graphic Display Data RAM (GDDRAM) for the screen which holds the bit pattern to be displayed. This memory area is organized in 6 banks (from 0 to 5). Each bank contains 84 columns/ segments (from 0 to 83) and each column can store 8 bits of data (from 0 to 7). It has eight pins.

Pin1 (RST) **:** This pin resets the module. It's an active low pin meaning; the display can be reset by pulling it low. This pin can also be connected to the microcontroller reset so that it will reset the screen automatically.

Pin2 (CE) **:** Chip Enable pin. This pin is used to select one of many connected devices sharing same SPI bus. It's an active low pin as well.

Pin3 (D/C) **:** Data/ Command pin. This pin tells the display whether the data it's receiving is a command or displayable data.

Pin4 (DIN) **:** Data input pin. This pin is a serial data pin for SPI interface. This is the input pin (MOSI) through which serial instructions are sent.

Pin5 (CLK) **:** This pin is a serial clock pin for SPI interface. All SPI modules require a common clock this clock source is supplied to this pin.

Pin6 (V_{cc}) **:** This pin is used to power the display the supply voltage is from 2.7 V to 3.3 V.

Pin7 (BL) **:** Backlight pin. This pin controls the backlight of the display. The ends of 10 kΩ potentiometer are connected to +5 V and ground then connected the middle pin to BL pin.

Pin8 (GND) **:** Ground pin.

The PCD8544 is a low power CMOS LCD controller (driver), designed to drive a graphic display 48 rows and 84 columns. All necessary functions for the display are provide in a single chip including on chip generation of LCD supply and bias voltages, resulting in a minimum of external components and low power consumption. The PCD8544 interfaces to microcontroller through a serial bus interface. Pin diagram and photograph of Nokia 5110 graphic LCD display are shown in Fig. (2.4) and Fig. (2.5).

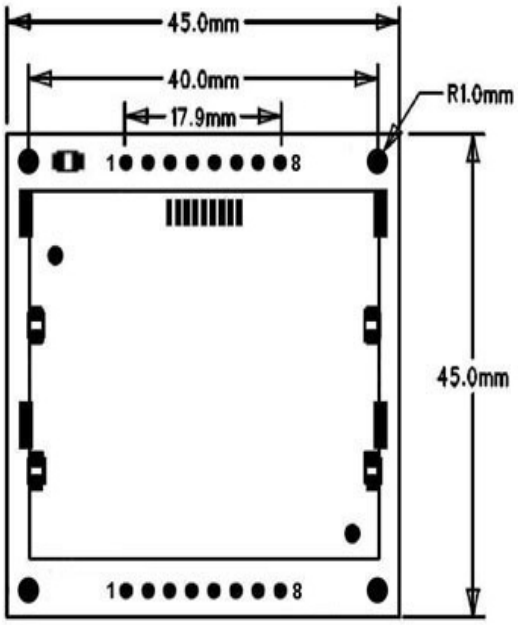
2.9 The Voltage Regulator

Voltage regulators are an essential part of most electronic hardware products. A voltage regulator is a system designed to automatically maintain a constant voltage level. It may use an electromechanical mechanism or electronic component. It keeps the voltage from a power supply within a range that is compatible with the other electrical components. Basically, there are two types of voltage regulators. They are linear voltage regulator and switching voltage regulator.

The most common series of voltage regulator is the 78×× series. The 78 series of voltage regulator produce positive voltages and use 4 digits (e.g. LM78××).

The last two digits of the number indicate output voltage for each voltage regulator. For example, a LM7805 ends with “05” means that the output voltage is 5 volts. Besides that, the first two digits are made to denote the series of regulators that output positive or negative voltage where LM78×× is the voltage regulators that output positive voltage and LM79×× is the voltage regulator that output negative voltage. A basic voltage regulator circuit consists of a regulator integrated circuit and filter capacitor.

The LM78×× is a common type of regulator known as a linear voltage regulator. Linear regulator such as the LM78×× become very popular because they are extremely easy to use: one of the pins of LM78×× is connected to the +V terminal of the battery. One of the pins is connected to the –V terminal of the battery. The third pin provides the + voltage output. In applications where a larger current is required, the LM78×× regulator IC can be used. This pin is compatible with a low power LM78×× should be used with a suitable heat sink in applications drawing more than a few hundreds of milli amperes. This IC is very flexible and is widely employed in all types of circuit like a voltage regulator. It is a three terminals device and mainly called input, output and ground. Photograph and pinout diagram of the LM78×× voltage regulator IC is shown in Fig. (2.6) below.



Pin	Name	Description
1	VCC	2.7 to 3.3V
2	GND	Ground
3	SCE	Chip enable (Active Low)
4	RES	Reset (Active Low)
5	D/C	Data/Command selection Low– Write command, High– Write data.
6	SDIN	Serial input
7	SCLK	Clock input
8	LED	Active High 2.7 to 3.2V

Fig. (2.4) Pin diagram of Nokia 5110 LCD



Fig. (2.5) Photograph of Nokia 5110 LCD

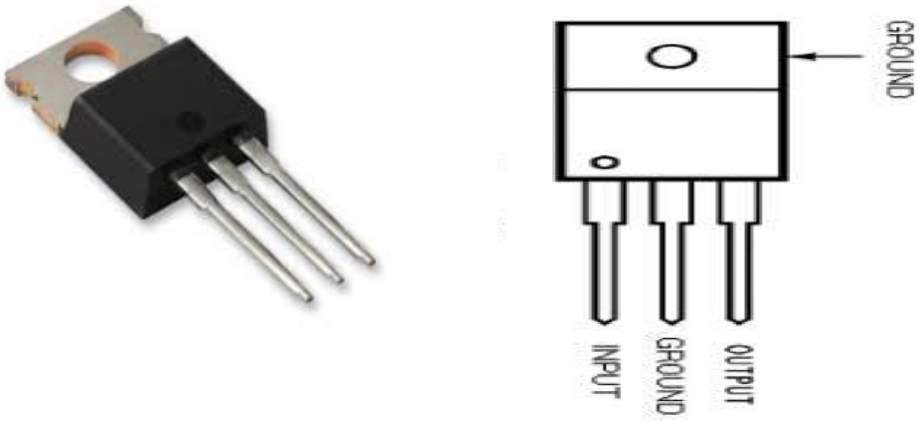


Fig. (2.6) The photograph and pinout diagram of the LM78×× voltage regulator

CHAPTER III

CONSTRUCTION OF CIRCUIT

3.1 Regulated Power Supply

The regulated power supply was constructed by using 7805IC, 7809IC, 12 V step-down transformer, DB107 p-n junction diode, a 2200 μF and a 100 μF capacitor. The DB107 p-n junction diode is used as full-wave bridge rectifier. It converts AC to DC. The input voltage must be DC voltage higher than the desired output voltage. ICs 7805 and 7809 are three terminals voltage regulators. The input voltage is given to pin 1 and pin 2 must be grounded. Pin 3 of the IC gives the output. The voltage regulator gives +5 V and +9 V. The circuit diagram of regulated power supply is shown in Fig. (3.1).

3.2 Interfacing temperature sensor (LM-35) to Microcontroller

The temperature sensor LM-35 IC has three pins. Two pins are used for the power supply and one pin for the analog output. It is low voltage IC which uses approximately +5 V DC of power. The output pin provides an analog voltage output that is linearly proportional to the Celsius (centigrade) temperature. Pin 2 gives an output of 10 mV per degree Celsius. So, to get the degree value in Celsius, all that must be done is to take the voltage value in degree of Celsius.

Pin 1 of the LM-35 goes into +5 V of the power supply. Pin 2 of the LM-35 is connected to pin 23 of Microcontroller. Pin 3 of the LM-35 goes into ground (GND) of the Microcontroller. LM-35 gives temperature output which is more accurately than thermistor output. Connection diagram of LM-35 to Microcontroller is shown in Fig. (3.2).

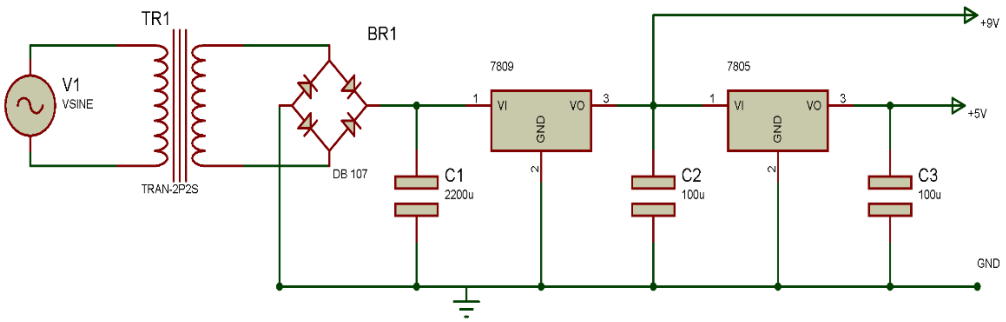


Fig. (3.1) Circuit diagram of power supply

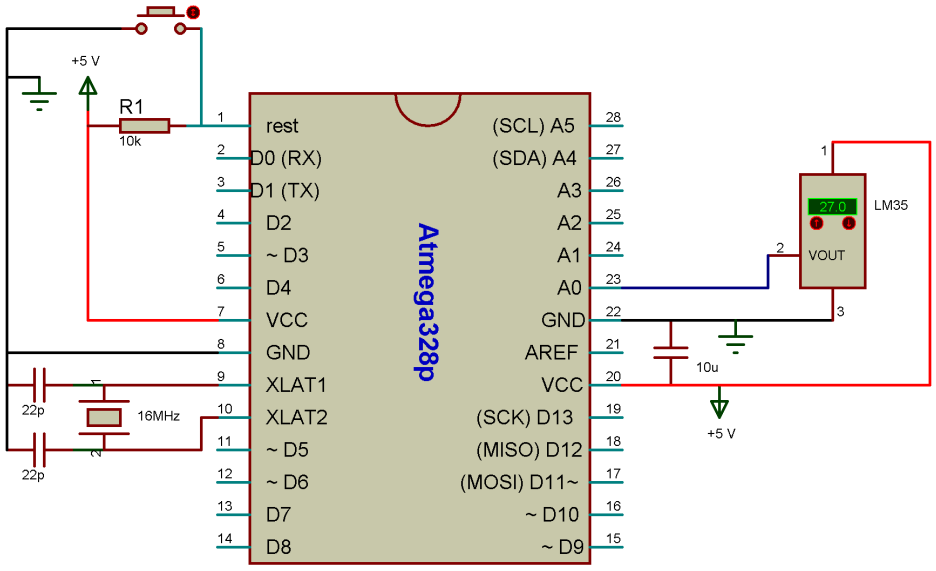


Fig. (3.2) Circuit diagram of interfacing LM-35 to microcontroller

3.3 Interfacing graphic LCD to Microcontroller

The Nokia 5110 LCD is very popular among microcontroller tinkerers. These modules are used on wide variety of applications that require some sort of interface or display data to user. The Nokia 5110 is interfaced to the microcontroller through a serial bus interface. Nokia 5110 LCD can display either the graphic or text. The Nokia 5110 LCD requires 3.3 V to operate, resistors must be used to convert the 5 V to 3.3 V. If the Nokia 5110 and microcontroller are operated without the resistors, then it would work but the life span of the LCD will decrease.

The circuit diagram of interfacing Nokia 5110 Graphic LCD to microcontroller is shown Fig (3.3). RST pin of LCD module is connected to digital pin 18 of the microcontroller. CE pin, DC pin, DIN pin of LCD module is connected to digital pin 17, 16, 15 of the microcontroller respectively. CLK pin of the LCD module is connected to digital pin 14 of the microcontroller. V_{cc} pin connects to the 3.3 V and GND pin connects to the GND pin of microcontroller. BL pin connects to the middle pin of 10 kΩ potentiometer and the other two pins of the potentiometer connect to the V_{cc} and GND pin of the microcontroller. The potentiometer is used to increase or decrease the backlight of LCD. So that the module of the backlight to be always high, it must be connected to 3.3 V pin of microcontroller or the backlight is not wanted, it must be connected to the GND pin of microcontroller. The signal connection for controlling operation of LCD is the serial format. Command mode and Data mode are the two modes that are used for communicating with LCD. The DC pin of the LCD is used for selecting the mode. If DC = 0, the data that is sent to LCD is command and if DC = 1, the data that is sent to LCD will be Data and it will be placed in DDRAM (Display Data RAM) to be displayed on LCD display.

3.4 Interfacing DC fan to microcontroller

A small driver circuit which consists of a resistor, C828 transistor (npn), 1N4007 diode and 10 V DC fan is used to interface the DC fan and microcontroller. Transistor is used to amplify the current so that full current (from the DC source- 9 V battery) can flow through this fan. The resistor is used to provide biasing to the transistor. The diode is used to prevent reverse current flow when the transistor is switched OFF. First of all, +9 V output pin of power supply must be connected to the positive terminal of DC fan. Pin 1 of transistor is emitter and it is connected to ground of microcontroller. Pin 2 of transistor is collector and it is connected to the negative terminal of DC fan. Pin 3 of transistor is base and it is connected to the microcontroller pin 14 through a 1 k resistor. Positive terminal of diode is connected to the negative terminal of DC fan. Negative terminal of diode is connected to the positive terminal of DC fan. Circuit diagram of interfacing DC fan to microcontroller is shown in Fig. (3.4).

3.5 Speed Control of DC Motor using PWM

Using PWM technique, the average value of the voltage that is applied to the DC Motor is controlled by turning the power on and off at a very high rate. The frequency of this switching will be in the order of few tens of kilo Hertz.

Now, the average voltage applied to the DC Motor will depend on the Duty Cycle of the PWM signal. Duty Cycle of a PWM signal is nothing but the ratio of the time for which the signal is ON or HIGH to the total time period of the signal. Duty Cycle is usually expressed in present and the represents different PWM signals of a 10 V supply with different duty cycles of 0%, 25%, 50%, 75% and 100% respectively. The speed of DC Motor will vary depending on the duty cycle of the PWM signal.

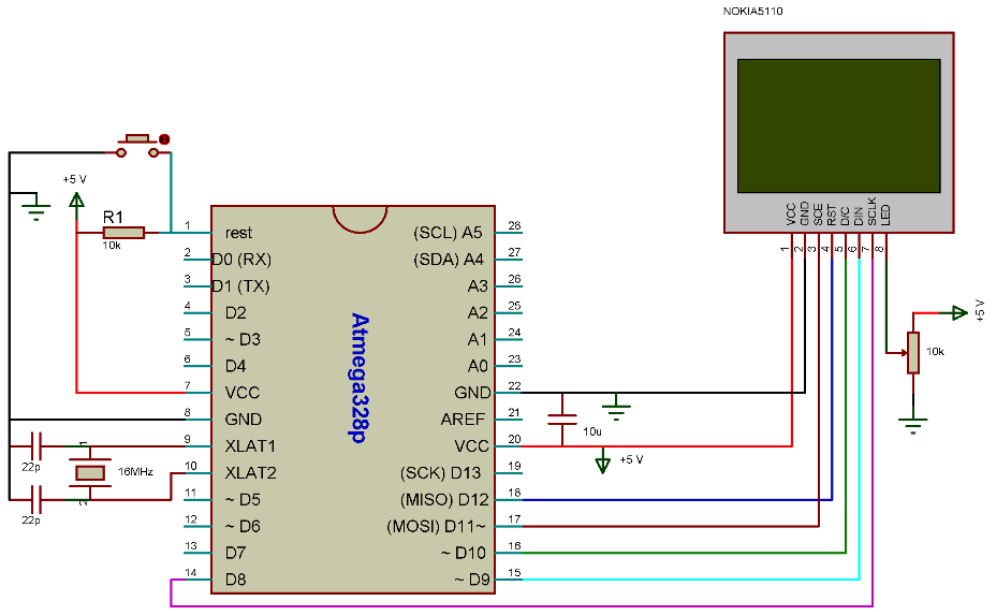


Fig. (3.3) Circuit diagram of interfacing LCD to microcontroller

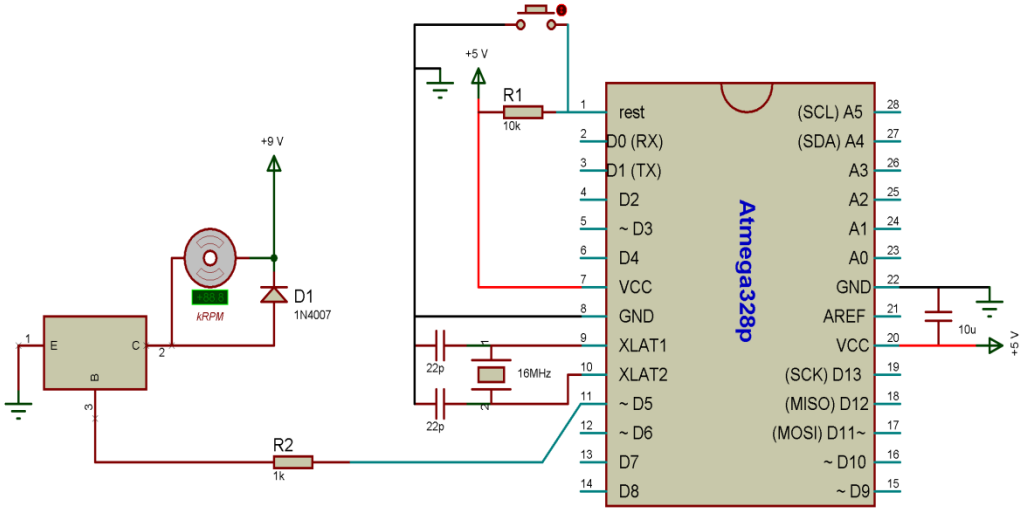


Fig. (3.4) Circuit diagram of interfacing DC fan to microcontroller

3.6 Working of the Whole Circuit

In this thesis, ATmega328 microcontroller, LM-35 temperature sensor, C828 transistor, 10 V DC fan and power supply are used. LCD shield is used to display the current temperature and speed of the fan. The temperature sensor senses the current temperature and the data is displayed on the LCD. This data is fed to the microcontroller. The microcontroller used has in built Pulse Width Modulation module which is used to control the speed of the fan by changing the duty cycle. If the current temperature is less than or equal to 20℃, the fan is OFF. If the current temperature is greater than or equal to 21℃, the fan is ON. When the current temperature is greater than 21℃ and less than 35℃, the fan speed is normal. The speed of fan is maximum when the current temperature is 35℃. The flow chart of temperature-controlled fan speed system is shown in Fig. (3.5). The zero percent duty cycle shows low speed. The 25 percent duty cycle shows slow speed and 50 percent duty cycle shows medium speed. The 75 percent duty cycle shows fast speed and very fast is shown by 100 percent duty cycle. According to the readings from the temperature sensor duty cycle is changed automatically thus controlling fan speed. The complete circuit diagram of temperature-controlled fan speed system is shown in Fig. (3.6).

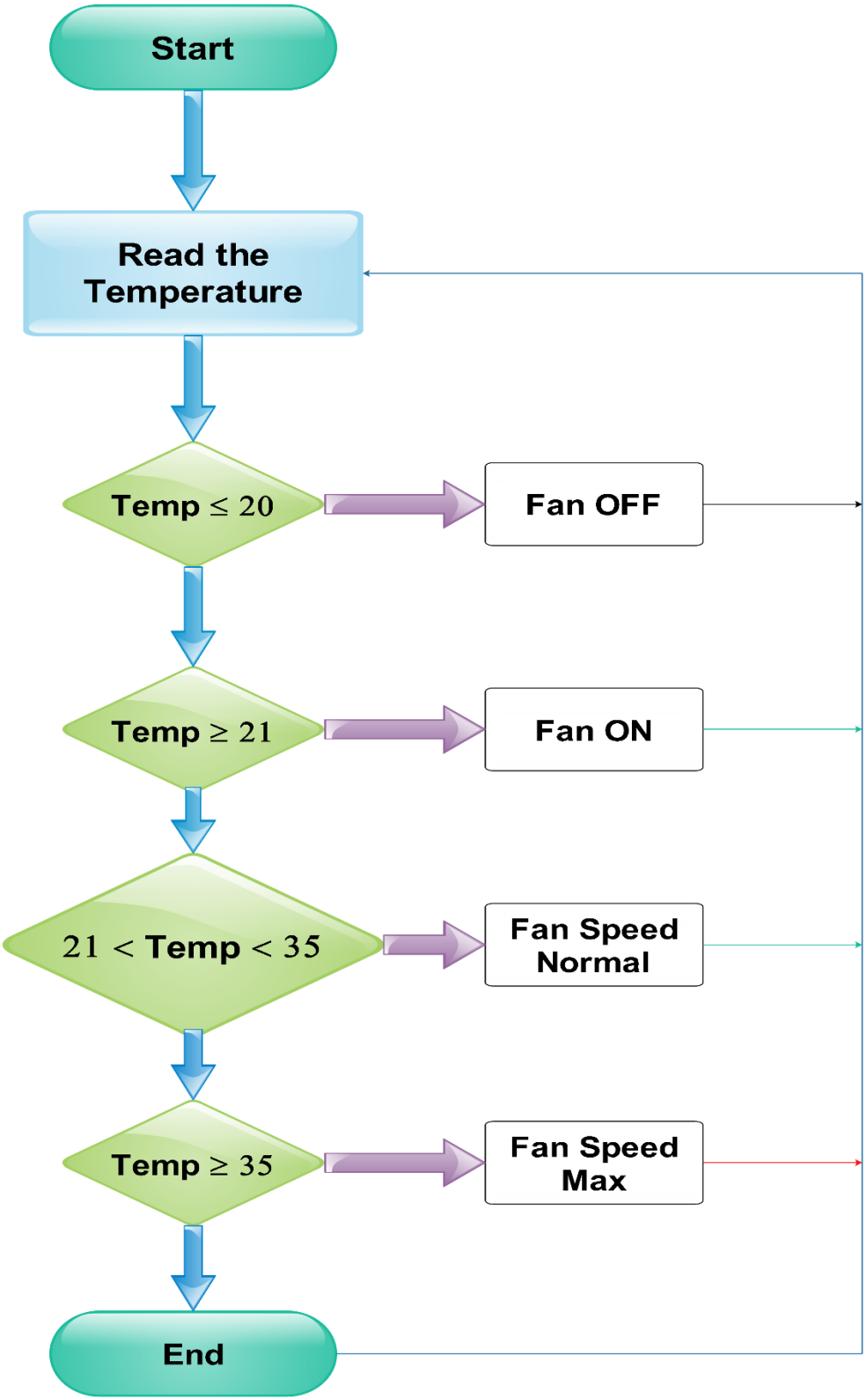


Fig. (3.5) Flow chart of temperature-controlled fan speed system

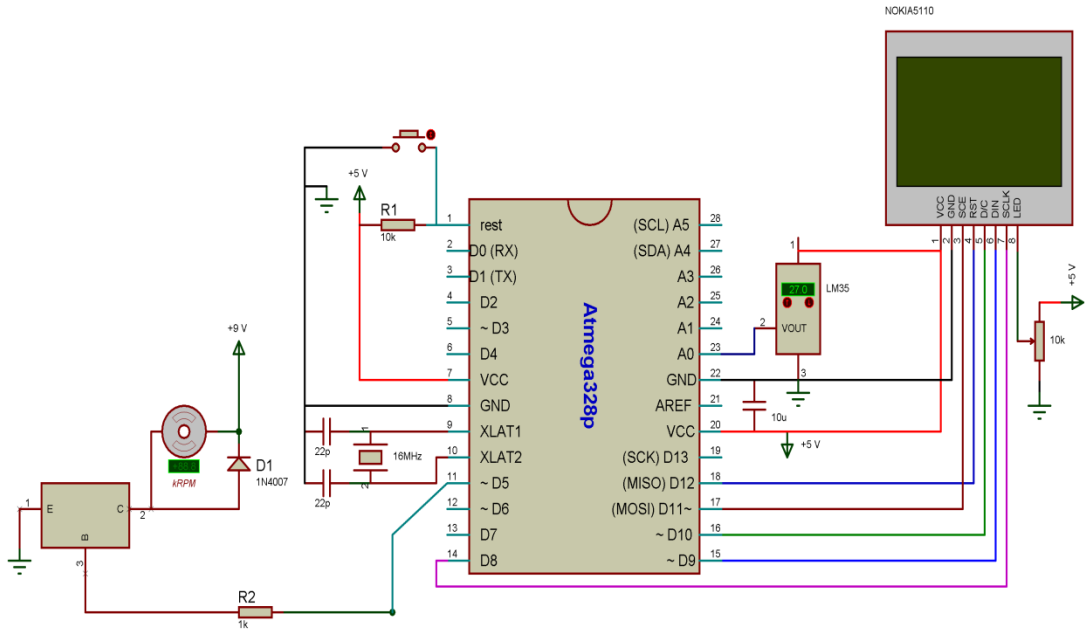


Fig. (3.6) Circuit diagram of temperature-controlled fan speed system

CHAPTER IV

RESULTS AND DISCUSSION

The speeds of DC motor can easily be varied. Therefore, they are utilized in applications where speed control, servo control or positioning needs exist and home appliance. Out of various control techniques available only two of them can be used to control DC motor. They are sensor control and sensor-less control. The idea behind this thesis is to control the speed of DC motor using microcontroller based on variation in temperature. Temperature controlled motor is an alternative way to deal with the speed of the motor. Microcontroller provides a suitable means of control and interface functions. The speed of motor depends on the room temperature and there is no need for regulating the speed manually.

In this system, the microcontroller is the main control unit. It accepts inputs from temperature sensor which allows for the measurement of the current room temperature, then the controller will give the action to maintain the required fan speed. The fan's speed will increase automatically if the temperature room increase. If the current temperature is greater than room temperature the fan speed will be turned on. If the current temperature is less than the room temperature, the fan speed will be turned off. The fan's speed will be change according to the room temperature. The motor speed has been varied from 0 to 100% which is displayed on LCD. Automatic fan speed regulator circuit is very useful in summer nights. As the temperature is high initially at night, the fan rotates with high speed. The lower the temperature, the slower the speed of fan rotates. The fan will stop when the room temperature getting cold. Photographs of constructed and running time complete microcontroller-based temperature-controlled fan speed are shown in Fig. (4.1) to Fig. (4.4).

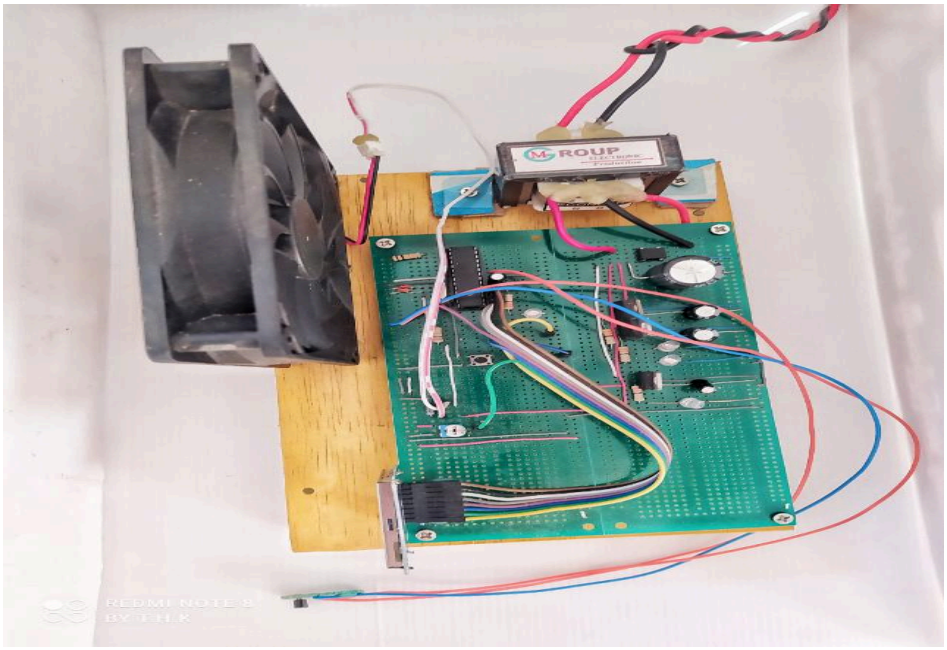


Fig. (4.1) Photograph of temperature-controlled fan speed circuit (Top View)

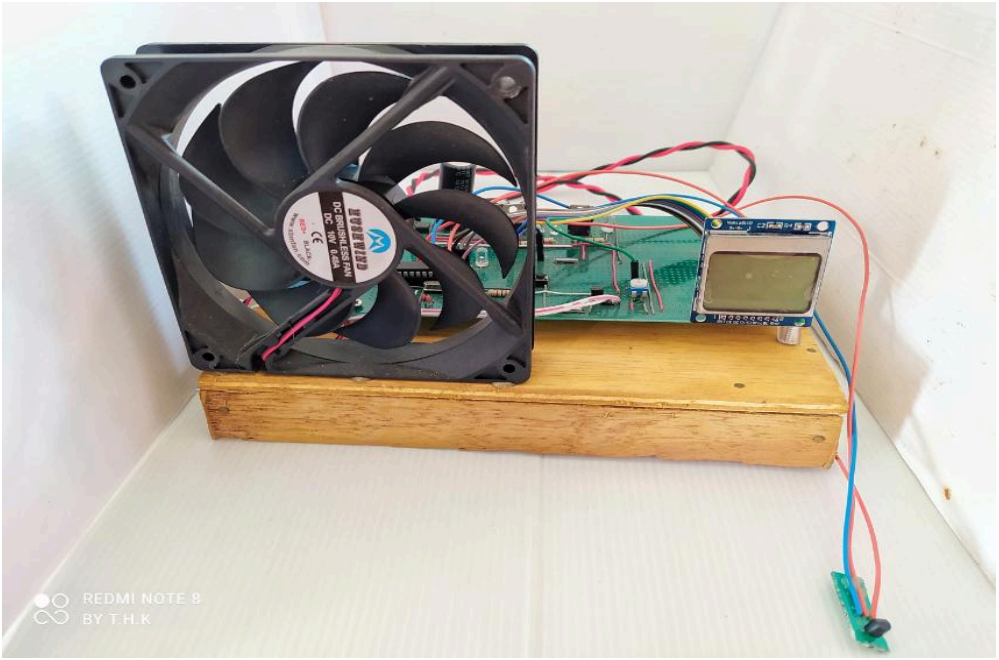


Fig. (4.2) Photograph of temperature-controlled fan speed circuit (Front View)

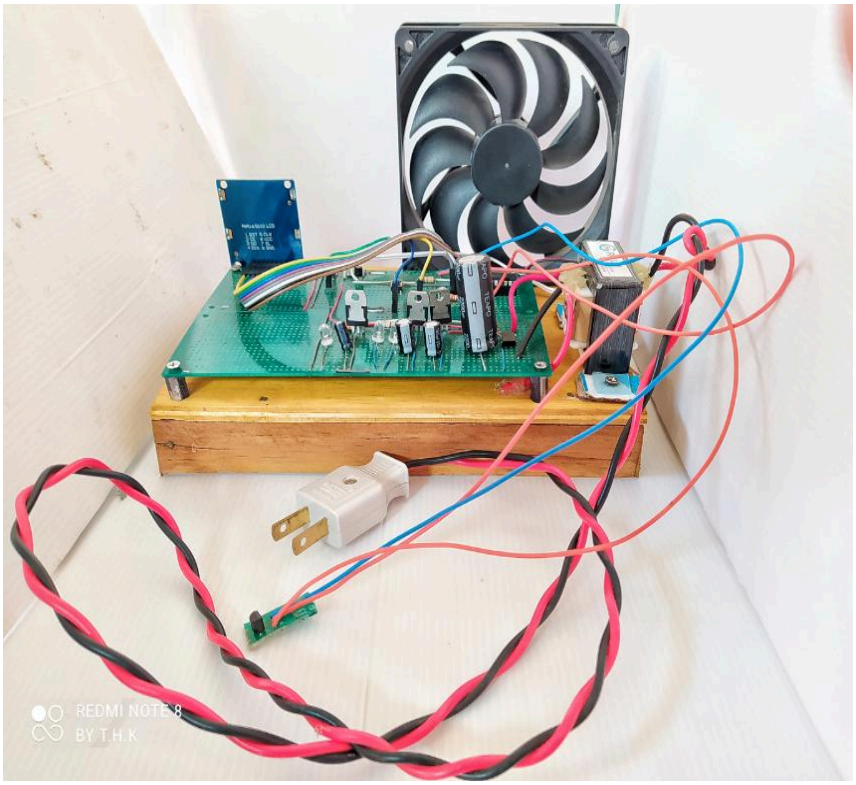


Fig. (4.3) Photograph of temperature-controlled fan speed circuit (Side View)

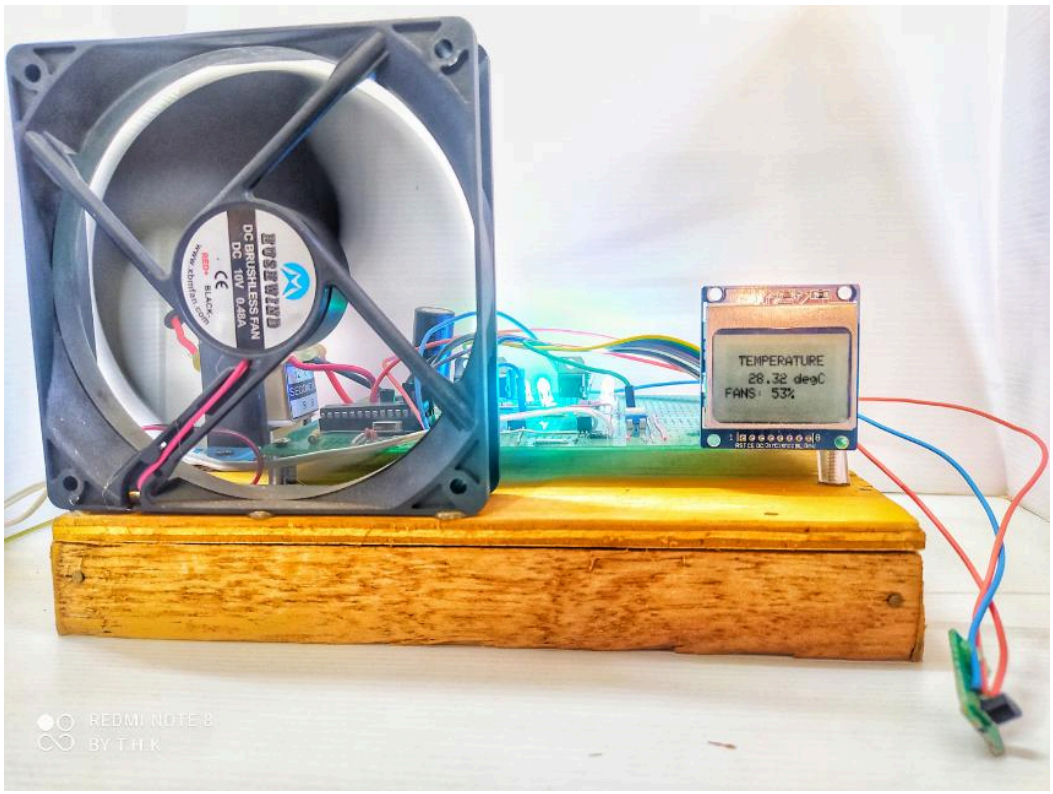


Fig. (4.4) Photograph of temperature-controlled fan speed at running time

CONCLUSION

In this thesis, the fan speed control system is designed and constructed to control the room temperature. The speed of DC motor is controlled by using Microcontroller and PWM technique. It is found to be that PWM technique is the best technique for controlling the fan's speed. The temperature sensor is used to measure the temperature of the room. Microcontroller is used to control the speed of DC motor. The speed of DC motor is varied with the room temperature. If the room temperature increase, the fan speed will increase automatically. There is no need for regulating the fan speed manually again and again. So, the fan speed control system is more comfortable than manual system. The required components are available in local market and low cost. The system which designed in this work was performed very well for any temperature change and can be classified as automatic control. It can be used in real life application.